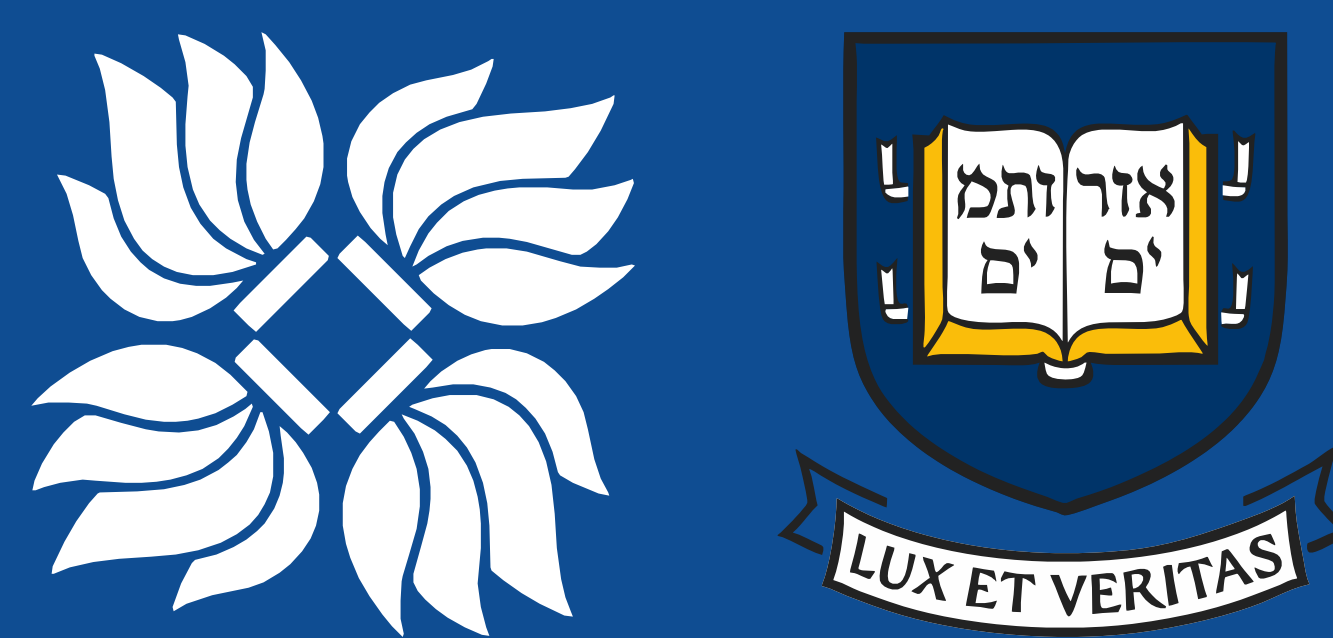




Chinese *gèng* and its norm-sensitive inference

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Overview

- Chinese *gèng* is an optional particle in comparison constructions
- Gèng* oftens brings a norm-sensitive inference (1)

(1) Naomi bǐ Mika (gèng) gāo.
Naomi STDD Mika (GÈNG) tall

≈ 'Naomi is (even) taller than Mika.'
Inference: 'Naomi is tall.'

- Discussions of the nature of this inference:
 - presupposition (e.g. LIU 2010, ZHANG 2020, CHEN & GREENBERG 2024)?
 - or implicature (e.g. GUO 2022, ZHANG & ZHANG-YUKUN 2025)?
- We argue that **the inference is an implicature**

Empirical Evidence

- English *even* brings a norm-sensitive inference (2) (GREENBERG 2018, 2022, ZHANG 2022)
- It is neither suspendable (see 3) nor cancellable (see 4)
 - suggests it's an implicature

(2) Naomi is even taller than [Mika]_F.
~~~~~**Presupp**: Mika is already tall.  
(3) I don't know whether Mika is tall or not, but Naomi is (#even) taller than Mika.  
(4) Naomi is even taller than [Mika]<sub>F</sub>. (#But in fact, Mika is {not tall / short}.)

- Chinese *gèng* behaves differently
  - The norm-sensitive inference is suspendable (5)
  - The norm-sensitive inference can be implicitly canceled (6)
  - The norm-sensitive inference can be explicitly canceled (7)

(5) **Suspended**  
Wǒ bù zhīdào Mika gāo bù gāo, dan Naomi kěndìng bǐ Mika  
1SG NEG know Mika tall NEG tall, but Naomi definitely STDD Mika  
(gèng) gāo.  
(GÈNG) tall

'I don't know whether Mika is tall or not, but Naomi is definitely taller.'

### (6) Implicitly canceled

a. Context: Naomi divides a cake into two parts to distribute between her and Mika. Mika complains that the part Naomi takes is significantly larger.  
Bù gōngpíng! [nǐ]<sub>F</sub> -de bùfèn (gèng) dà.  
NEG fair [2SG]<sub>F</sub> -GEN part (GÈNG) big

'It's unfair! Your part is bigger than mine.'  
~~~~~  
~~~~~ My share is too small (below norm).

b. Zhè jiā diàn-de dàngāo [yǐqián]<sub>F</sub> (gèng) hǎochī.  
DEM.PROX CLF store-GEN cake [before]<sub>F</sub> (gèng) delicious

'This store's cakes used to be more delicious.'  
~~~~~  
~~~~~ not delicious now (below norm).

### (7) Explicitly Canceled

Naomi zuò-de bù hǎo, dàn yě méi rén zuò-de bǐ tā  
Naomi do-LNK NEG good but also NEG.be person do-LNK STDD 3SG  
(gèng) hǎo.  
(gèng) good

'Naomi didn't do well, but no one did better than her.'

~~~~~  
~~~~~ ~~Naomi did well.~~

## Analysis

- We propose that:
  - Gèng*-less comparison ≈ English explicit comparison
  - Gèng*-ful comparison ≈ English implicit comparison
- Gèng*-less (8a) and *gèng*-ful (8b) comparison share the same truth-conditions
  - Yet only *gèng*-less comparison (8b) gives rise to a norm sensitive inference (see also SAWADA 2009)

(8) a. Without *gèng* **Explicit comparison**  
[1] ≈ Naomi is taller than Mika.  
b. With *gèng*: **Implicit**  
[1] ≈ Compared to Mika, Naomi reaches a new height.

- Supported by *gèng*'s incompatibility with numerical differences (9)

(9) Naomi bǐ Mika (\*gèng) gāo wǔ límǐ.  
Naomi STDD Mika (\*GÈNG) tall five centimeter  
  
Without *gèng*: ✓ 'Naomi is 5 cm taller than Mika.'  
With *gèng*: \*\*\*numerical difference\*\*\*

- The norm-sensitive inference can be derived as a Gricean or RSA implicature (e.g. LASSITER & GOODMAN 2017)

(10) Meaning space: { $M_1, M_2, M_3$ }  
 $M_1$ : HT( $m$ ) < HT( $n$ ) < POS  
 $M_2$ : HT( $m$ ) < POS < HT( $n$ )  
 $M_3$ : POS < HT( $m$ ) < HT( $n$ )  
(11) Utterance space: { $U_{POS}, U_{CMPR}, U_{GÈNG}$ }  
 $U_{POS}$ : Naomi is tall  
 $U_{CMPR}$ : Naomi bǐ Mika gāo  
 $U_{GÈNG}$ : Naomi bǐ Mika gèng gāo

- A pragmatic speaker selects the utterance that is maximally informative for identifying the targeted meaning from the meaning space, and a pragmatic listener makes use of this utility estimation to update probability distribution (Table 1)
- If  $M_2$ ,
  - $U_{POS}$  is true, maximally informative (not less informative than  $U_{CMPR}$  or  $U_{GÈNG}$ ), and most economic
  - A pragmatic speaker thus chooses  $U_{POS}$
- If  $M_1$ ,
  - $U_{POS}$  is false and violates the Maxim of Quality
  - Both  $U_{CMPR}$  and  $U_{GÈNG}$  are true, but  $U_{CMPR}$  is more economic
  - A pragmatic speaker thus prefers  $U_{CMPR}$
- Therefore,  $U_{GÈNG}$  is optimal only under  $M_3$ , because its lower economy has to be offset by greater informativeness (in addressing the QUD 'how tall Naomi is')

|       | $U_{POS}$        | $U_{CMPR}$       | $U_{GÈNG}$       |
|-------|------------------|------------------|------------------|
| $M_1$ | False            | True (preferred) | True             |
| $M_2$ | True (preferred) | True             | True             |
| $M_3$ | True             | True             | True (preferred) |

Table: Deriving the norm-sesntive inference

## References

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